Discovering Unusual Customer Purchasing Behavior

PROJECT PROPOSITION

Data Mining Project

CSCA 5522

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Abstract

- Investigates unusual customer purchasing behavior in UK online retail transactions
- Invoice-level dataset: each transaction’s own features + aggregated customer-level behavior
- Goal: discover/characterize behavioral segments, then find transactions & customers that don’t fit any of them
- Not a predictive-modelling exercise; clustering is a means, not the end
- Main axes of interest: cancellation behavior and order total

Motivation: Business Value

- Identify high-value customers
- Identify segments associated with frequent cancellations
- Detect emerging changes in purchasing patterns
- Supports retention, marketing, and operational decisions

Objective

- Identify and characterize purchasing behavior that differs substantially from the major segments
- Trace those outliers back to the customers who produced them
- Example: if 3 well-defined clusters emerge - which transactions/customers fit none of them?

Related Work

- Customer Segmentation & Retention Strategy (Kaggle) - segments as end product for retention
- Online Retail Cohort Analysis (Kaggle, same dataset) - descriptive cohorts/trends, not segmentation+outliers
- Enhanced RFM / RFMT (Kaggle) - adds Inter-Purchase-Time to RFM; segmentation is the end goal there
- Recommendation Systems (Kaggle) - association rules & collaborative filtering, different question entirely
- This project differs: segmentation is a baseline, not the deliverable - the deliverable is what doesn't fit

How this work differs from the existing work:

There is plenty of work related to retail data. It can largely be divided into three main categories: customer segmentation, prediction tasks (e.g., customer churn, likelihood of cancellation), and recommendation systems.

I chose to do something distinct - something with a deeper affinity to the "Data Mining" discipline itself: intentionally avoiding pure classification, regression, and clustering as an end goal, and instead focusing on discovering unknown patterns and analyzing the data.

Proposed Work: Feature Engineering

- Filter to UK, drop guest transactions
- Build customer-level features:
  - Timing: day of week, hour, avg. shopping trips per month
  - Spending: avg/min/max/std of transaction totals
  - Product: avg/min/max/std of items purchased, most frequent products
  - Cancellations: % of cancelled transactions
- Join customer-level features back onto invoice-level records
- Feasibility: ~50 engineered features, ~40,500 invoices, ~5,400 customers

Proposed Work: Analysis Flow

- Explore segmentation using engineered behavioral features
- Characterize the common patterns found in each segment
- Identify transactions/customers whose behavior differs significantly from the majority or from any segment
- Particular focus: spending, purchase composition, cancellation behavior

Evaluation: Clustering

- Baseline: KMeans (quick, cheap, first impression - cannot produce outliers)
- Working model: DBSCAN (produces outliers - the "working horse" of this research)
- Two distance metrics evaluated: euclidean and cosine
  - DBSCAN supports cosine natively; "cosine KMeans" approximated via L2-normalization + euclidean KMeans
- Evaluated on: silhouette score (direction, not verdict), stability under hyper-parameter changes, interpretability
- Target: stable, 2-3 interpretable clusters, manageable/interpretable outliers

Evaluation: Characterizing Differences

- Question: how does cluster A differ from cluster B? How do outliers differ from inliers?
- Raw mean differences aren’t comparable across features on different scales
- Solution: Cohen’s d - standardized mean difference, comparable across features
- Take the top 5-10 features with |Cohen’s d| > 0.8 to characterize/define each population

Timeline

- **Week 1:** EDA, preprocessing, and feature engineering
- **Week 2:** Unusual purchasing behavior analysis
- **Week 3:** Report preparation

References

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